18.6.4 Osmoregulation

The term osmoregulation was coined by **Hober** in 1902. Osmoregulation is the process by which an organism regulates the water balance in its body and maintains the homeostasis of the body. It includes controlling excess water loss or gain and maintaining the fluid balance and the osmotic concentration, that is, the concentration of electrolytes. It ensures that the fluids in the body do not get too diluted or concentrated.

Organisms are divided into two types based on osmoregulation. They are Osmoconformers and Osmoregulator.

Osmoconformers

These organisms try to maintain the osmolality of their body matching with their surroundings. Most of the invertebrates, marine organisms are osmoconformers.

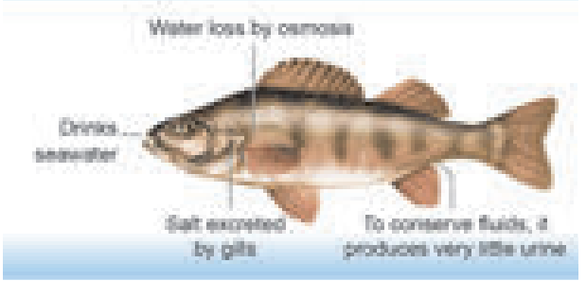


Figure 18.13 Osmoregulation by a marine fish

Osmoregulators

These organisms maintain their internal osmolality, which can be extremely different from that of the surrounding environment, through physiological processes

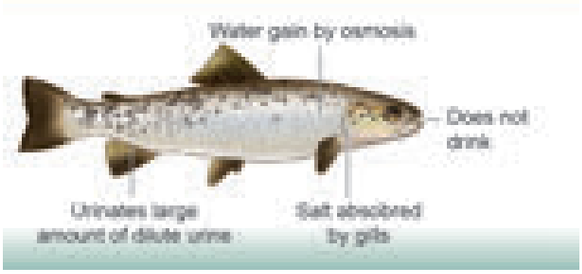


Figure 18.14 Osmoregulation by a freshwater fish

18.6.5 Cellular respiration

Cellular respiration is the process by which organisms break down glucose into a form that the cell can use as energy. This energy is then made available to living cells in the form of **ATP**. Cellular respiration takes place in the cytoplasm and mitochondria of the cells. The cellular respiration is classified into two types: **aerobic** respiration and **anaerobic** respiration.

a. Aerobic respiration

In this type of respiration, the food substances are completely oxidized into H_2O and CO_2 with the release of energy. It requires atmospheric oxygen and all higher organisms respire aerobically. This reaction releases a large amount of energy.

Glucose + Oxygen → Carbon dioxide + Water + Energy

b. Anaerobic respiration

In this type of respiration, partial oxidation of food takes place and the organisms release energy in the absence of oxygen. This type of respiration occurs in organisms like yeast. Ethyl alcohol or lactic acid and carbon dioxide are the by-products of this process. This reaction releases very little energy because glucose is not completely oxidized.

Table 18.2 Differences between aerobic and anaerobic respiration

Aerobic	Anaerobic
Aerobic respiration takes place in the presence of oxygen.	Anaerobic respiration takes place in the absence of oxygen.
The end products of aerobic respiration are carbon dioxide and water.	The end products of anaerobic respiration are CO_2 and ethanol or lactic acid.
Common in all higher plants and animals.	Common in certain micro organisms and human muscle cell.

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